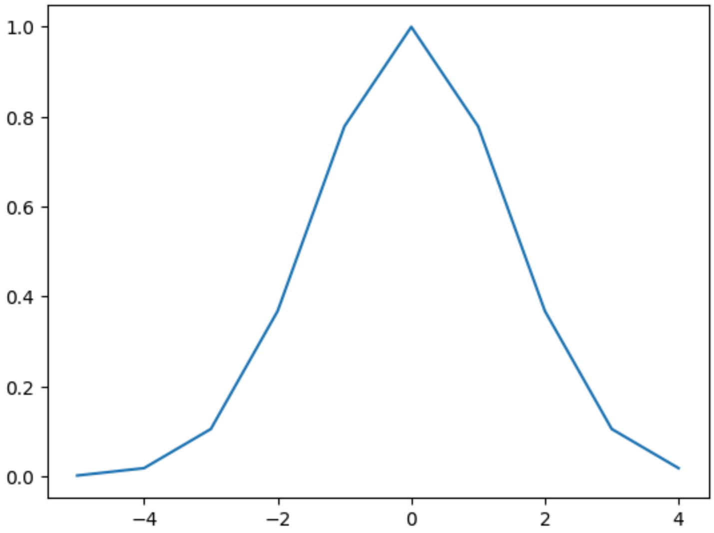
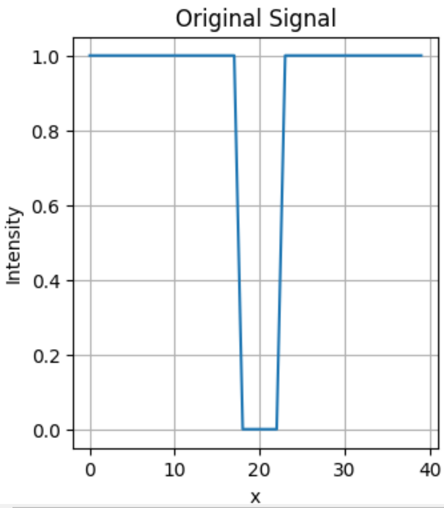
Graded Task 2

```
import numpy as np
import matplotlib.pyplot as plt


def laplacian_of_gaussian(sigma, size):
    # todo: compute the laplacian of gaussian
    gaussian = 1 / (sigma * np.sqrt(2 * np.pi)) * np.exp(-np.arange(-size // 2 + 1, size // 2 + 1)**2 / (2 * sigma**2))
    laplacian = np.diff(gaussian, n=2)
    return laplacian


# Testing it out
sigma = 2
x = np.arange(-5, 5)
gaussian = np.exp(-x**2 / (2 * sigma))
plt.plot(x, gaussian)
```

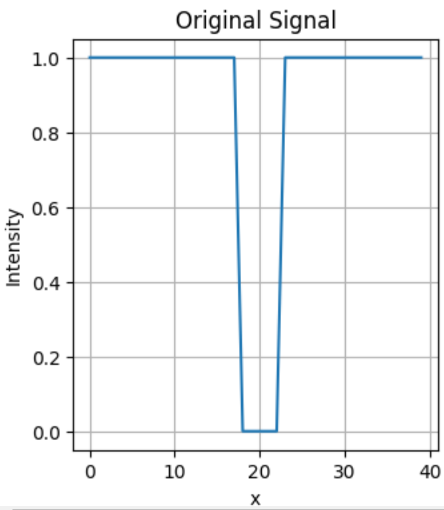
```
[<matplotlib.lines.Line2D at 0x7b42a5f2ee60>]
```



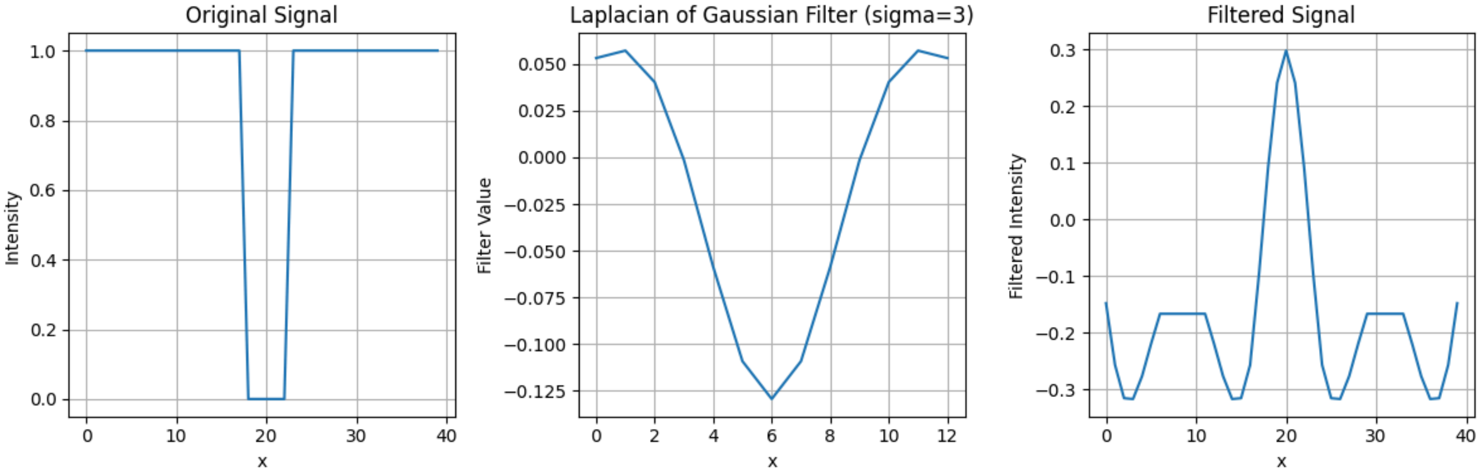

```
# Create a 1D blob signal
blob_size = 5
blob_center = 20
signal_length = 40
signal = np.ones(signal_length)
signal[blob_center - blob_size // 2:blob_center + blob_size // 2 + 1] = 0
plt.figure(figsize=(12, 4))
plt.subplot(1, 3, 1)
plt.plot(signal)
plt.title("Original Signal")
plt.xlabel("x")
plt.ylabel("Intensity")
plt.grid(True)
```




```
# Apply the LoG to the above signal at sigma = 5 and plot results
# use np.convolve


def laplacian_of_gaussian(sigma, size):
  gaussian = np.exp(-np.arange(-size // 2 + 1, size // 2 + 1)**2 / (2 * sigma**2)) / (sigma * np.sqrt(2 * np.pi))
  laplacian =   sigma**2 *np.diff(gaussian, n=2)
  return laplacian


# Create a 1D blob signal
blob_size = 5
blob_center = 20
signal_length = 40
signal = np.ones(signal_length)
signal[blob_center - blob_size // 2:blob_center + blob_size // 2 + 1] = 0
plt.figure(figsize=(12, 4))
plt.subplot(1, 3, 1)
plt.plot(signal)
plt.title("Original Signal")
plt.xlabel("x")
plt.ylabel("Intensity")
plt.grid(True)
```



```
# Apply the LoG to the above signal at various sigma values and plot the result.
# use np.convolve
sigma = 3
size = 15
log_filter = laplacian_of_gaussian(sigma, size)
filtered_signal = np.convolve(signal, log_filter, mode='same')

# Plot the original signal, the LoG filter, and the filtered signal
plt.figure(figsize=(12, 4))
plt.subplot(1, 3, 1)
plt.plot(signal)
plt.title("Original Signal")
plt.xlabel("x")
plt.ylabel("Intensity")
plt.grid(True)

plt.subplot(1, 3, 2)
plt.plot(log_filter)
plt.title(f"Laplacian of Gaussian Filter (sigma={sigma})")
plt.xlabel("x")
plt.ylabel("Filter Value")
plt.grid(True)

plt.subplot(1, 3, 3)
plt.plot(filtered_signal)
plt.title("Filtered Signal")
plt.xlabel("x")
plt.ylabel("Filtered Intensity")
plt.grid(True)

plt.tight_layout()
plt.show()
```




```
# Apply LoG at various sigma values and plot the response values at blob center against various sigma values
```

```
# Apply the LoG to the above signal at various sigma values and plot the result.
# use np.convolve
response = []

sigmas = np.linspace(1, 10)

for sigma in sigmas:
  log = laplacian_of_gaussian(sigma, 21)
```

```
    filtered_signal = np.convolve(signal, log, mode = "same")
    response.append(filtered_signal[20])


# Plotting the graph
plt.plot(response)
```

```
[<matplotlib.lines.Line2D at 0x7b429582e860>]
```